UNIVERSITY OF AGRICULTURAL SCIENCES SOIL ANALYSIS WING, HASSAN

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA05	**Report Date:**	19-Oct-2026
Location / District:	Belur Taluk, Hassan District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Red Sandy Clay Loam	**Target Agro-Zone:**	Karnataka - Hassan Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**225.0**	kg/ha	280.0 - 560.0	Medium (Normal)
Available Phosphorus	**P**	**34.0**	kg/ha	23.0 - 56.0	Medium (Adequate)
Available Potassium	**K**	**215.0**	kg/ha	140.0 - 330.0	Medium (Balanced)
Soil Reaction (pH)	**pH**	**6.40**	pH scale	6.5 - 7.5	Slightly Acidic (Favorable)
Electrical Conductivity	**EC**	**0.28**	dS/m	< 1.00	Normal (Low Salt)
Organic Carbon	**OC**	**0.55**	%	0.50 - 0.75	Medium (Satisfactory)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 225.0 kg/ha [Medium (Normal)]

Available Phosphorus (P): 34.0 kg/ha [Medium (Adequate)]

Available Potassium (K): 215.0 kg/ha [Medium (Balanced)]

Soil Reaction (pH): 6.40 [Slightly Acidic (Favorable)]

Electrical Conductivity (EC): 0.28 dS/m [Normal (Low Salt)]

Organic Carbon (OC): 0.55 % [Medium (Satisfactory)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Moderately fertile transition zone red loam with balanced moisture retention and favorable physical properties. Balanced nutrient ratios across primary macronutrients.

General Nutrient Management Guidance: Maintain balanced NPK fertilization (4:2:1 ratio). Apply 10 t/ha compost blended with Trichoderma to preserve soil aggregate stability and protect roots from soil-borne pathogens. Optimal for potato, maize, vegetables, and ginger.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*